



Graphs in Machine Learning

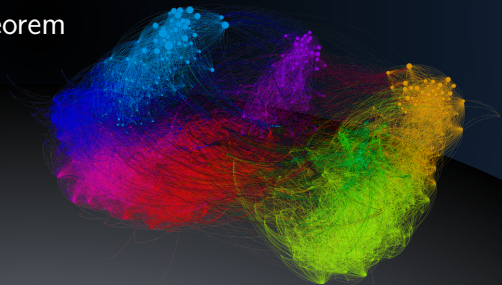
Google PageRank: Core Algorithm

Steady State and Perron's Theorem

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Partially based on material by: Andreas Krause,
Branislav Kveton, Michael Kearns



Success story #2 Google PageRank

Google matrix: $G = (1 - p)M + p \cdot \frac{1}{N} \mathbb{1}_{N \times N}$, where $p = 0.15$

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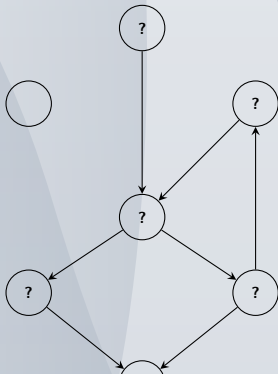
$\mathbf{G}$ is **stochastic** why? What is G_{aa} for any a ? We look for $\mathbf{G}\mathbf{v} = \mathbf{1} \times \mathbf{v}$,
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Perron's theorem: Such v exists and it is **unique** if the entries of G are positive.

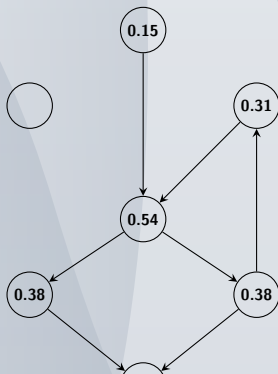


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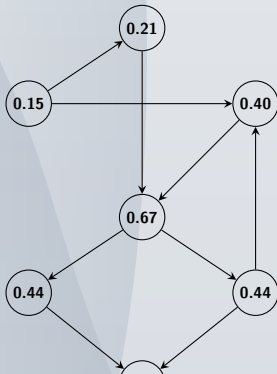


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<https://misovalko.github.io/mva-ml-graphs.html>